

Spherical Model of a Ferromagnet

Phase Transitions and Critical Phenomena (P455): Term Paper

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Abstract

This paper discusses the spherical model of the ferromagnet, a truncation method of understanding critical phenomena and phase transitions in statistical mechanics. First developed in 1952 by Berlin and Kac, the spherical model liberalizes the Ising model constraint by permitting continuous spin values, not discrete ± 1 or ± 1 . This adjustment renders the model solvable analytically in arbitrary dimensions, a singular tool for investigating critical behavior in systems of different dimensionalities. Specifically, the spherical model gives an exact solution to ferromagnetic systems, allowing one to investigate critical phenomena like phase transitions, specific heat, and magnetization. We then recapitulate the Gaussian model as an antecedent to the spherical model, prior to the detailed analysis of the partition function and free energy. The resulting magnetization and specific heat expressions are calculated via saddle-point approximations and integration methods. Additionally, the effect of an applied external homogeneous magnetic field is included in the model, resulting in a new Hamiltonian and new formulas for free energy and magnetization. The article is finished with an outline of the importance of the model in contemporary research, especially in the framework of quantum versions of the spherical model and their applications in quantum information and many-body physics. The spherical model remains a vital part of phase transition understanding in both classical and quantum systems and provides important insights into the nature of physical system critical phenomena.

1 Introduction

In the study of phase transitions, especially how a material like a ferromagnet spontaneously develops magnetization below its critical temperature, has always been a challenging problem in statistical mechanics. Numerous efforts have been made to create models that can accurately describe this behavior, but many of these models are too complex to solve exactly, especially in three dimensions. A notable example is the 3 dimensional Ising model, which remains unsolved to this day. That is where the Spherical Model of a Ferromagnet[1], proposed by T. H. Berlin and M. Kac in 1952, comes into the play. This model simplifies the problem by relaxing the strict constraints on individual spins, allowing for an exact solution in arbitrary dimensions, allowing us to explore critical phenomena and phase transitions in a three dimensional setting.

In the classic Ising model, each spin can only take on discrete values of $+1$ or -1 . Berlin and Kac relaxed this constraint by allowing the spins to take continuous values by replacing it with a global constraint on the sum of the squares of the spins. This made the model mathematically tractable in arbitrary dimensions, and exhibited a phase transition in all dimensions greater than two. Apart from the exact solution, the model proved to be a valuable tool for understanding Bose-Einstein condensation[2] and also in the study of critical phenomena in systems of finite thickness[3].

2 The Gaussian Model

The Spherical Model was introduced as a modification of the Gaussian model by Berlin and Kac. The Gaussian model is a continuous version of the Ising model where the spins can take any real value, and the Hamiltonian is given by $H = -J \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j$ where J is the interaction strength, and the sum is over nearest neighbor pairs. Each spin ϵ_i is drawn from a Gaussian distribution with zero mean and unit variance. That is, the probability of finding the i^{th} spin with between ϵ_i and $\epsilon_i + d\epsilon_i$ is given by:

$$P(\epsilon_i)d\epsilon_i = \frac{1}{\sqrt{2\pi}} e^{-\frac{\epsilon_i^2}{2}} d\epsilon_i \quad (1)$$

This yields the partition function as:

$$Z = \frac{1}{(2\pi)^{N/2}} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \left(\prod_{i=1}^N d\epsilon_i \right) \exp \left(-\frac{1}{2} \sum_{i=1}^N \epsilon_i^2 + K \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j \right) = \exp \left(-\frac{1}{2} \sum_{p=1}^N \ln(1 - 2K\lambda_p) \right) \quad (2)$$

where $K = \beta J$, and λ_p for different lattices are given in the table below:

Lattice	λ_p
1-dimensional chain	$2\cos\left(\frac{2\pi(p-1)}{N}\right)$
2-dimensional square lattice	$2\cos\left(\frac{2\pi(p-1)}{N}\right) + 2\cos\left(\frac{2\pi n_1(p-1)}{N}\right)$
3-dimensional cubic lattice	$2\cos\left(\frac{2\pi(p-1)}{N}\right) + 2\cos\left(\frac{2\pi n_1(p-1)}{N}\right) + 2\cos\left(\frac{2\pi n_1 n_2(p-1)}{N}\right)$

Table 1: Values of λ_p for different lattices

Here, n_1 and n_2 are the number of sites along the two dimensions of the 2D lattice, and $N = n_1 n_2$ is the total number of sites. Similarly, for the 3D lattice, n_1 , n_2 , and n_3 are the number of sites along the three dimensions, and $N = n_1 n_2 n_3$ is the total number of sites.

The free energy per spin in the thermodynamic limit can be found using the partition function as:

$$\psi = \frac{k_B T}{2} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{p=1}^N \ln(1 - 2K\lambda_p) \quad (3)$$

On substituting the values of λ_p from Table 1, we find that the free energy per spin for the 1D, 2D, and 3D lattices are in agreement with the form of the free energy obtained from the Ising model. Despite this, the free energy becomes complex for $K > K_c = \frac{1}{4n}$, where n is the dimensionality of the lattice. This indicates that the Gaussian model is not well defined below a critical temperature. So, we introduce the spherical model, which retains all important features of the Gaussian model and is valid for all temperatures.

3 The Spherical Model

A spin configuration $\epsilon = (\epsilon_1, \epsilon_2, \dots, \epsilon_N) \in \mathbb{R}^N$ is an allowed configuration in the Ising model if $\epsilon_i = \pm 1$ for all $1 \leq i \leq N$. That is, ϵ lies on the vertices of an N -dimensional hypercube of side length 2 units centered at the origin. Consider a sphere of radius $\sqrt{N}$ centered at the origin in $\mathbb{R}^N$, $S_{N-1}(\sqrt{N}) = \{\epsilon \in \mathbb{R}^N : \sum_{i=1}^N \epsilon_i^2 = N\}$. Clearly, all allowed configurations of the Ising model lie on this sphere. The spherical model allows every point on $S_{N-1}(\sqrt{N})$ to be an

allowed configuration.

The spherical constraint can be modelled in the partition function as follows:

$$Z = \frac{\Gamma(N/2)}{2\pi^{N/2}N^{(N-1)/2}} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \left(\prod_{i=1}^N d\epsilon_i \right) \exp \left(K \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j \right) \delta \left(N - \sum_{i=1}^N \epsilon_i^2 \right) \quad (4)$$

3.1 Evaluating the partition function and the free energy

The spherical constraint can be modelled in the partition function as follows:

$$Z = \frac{\Gamma(N/2)}{2(\pi)^{N/2}N^{(N-1)/2}} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \left(\prod_{i=1}^N d\epsilon_i \right) \exp \left(K \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j \right) \delta \left(N - \sum_{i=1}^N \epsilon_i^2 \right) \quad (5)$$

$$= \frac{\Gamma(N/2)}{2(\pi)^{N/2}N^{(N-1)/2}} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \left(\prod_{i=1}^N d\epsilon_i \right) \exp \left(K \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j \right) \frac{1}{2\pi i} \int_{-i\infty}^{i\infty} ds \exp \left\{ s \left(N - \sum_{i=1}^N \epsilon_i^2 \right) \right\} \quad (6)$$

We now wish to interchange the ϵ and s integrals, but the resulting inner integral is not convergent. We analytically continue the expression of the Hamiltonian to $\mathbb{R}^N$ instead of just the unit sphere. And we re-write

$$K \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j = N\alpha_0 - \alpha_0 \sum_{i=1}^N \epsilon_i^2 + K \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j \quad (7)$$

By choosing α_0 to be a sufficiently large positive number, we can ensure that the coefficient of the quadratic term of ϵ in the exponent is negative. This solves our previous problem of the integral diverging.

$$Z = \frac{\Gamma(N/2)}{2(\pi)^{N/2}N^{(N-1)/2}} \frac{1}{2\pi i} \int_{-i\infty}^{i\infty} ds \int_{\mathbb{R}^N} d^N \epsilon \exp \left\{ N(s - \alpha_0) - (s - \alpha_0) \sum \epsilon_i^2 + K \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j \right\} \quad (8)$$

We can now shift the contour that we are integrating over to the right by α_0 to get

$$Z = \frac{\Gamma(N/2)}{2(\pi)^{N/2}N^{(N-1)/2}} \frac{1}{2\pi i} \int_{\alpha_0 - i\infty}^{\alpha_0 + i\infty} ds \exp\{Ns\} \int_{\mathbb{R}^N} d^N \epsilon \exp \left\{ -s \sum \epsilon_i^2 + K \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j \right\} \quad (9)$$

$$(10)$$

We can rewrite the interaction term in the Hamiltonian in a more tractable way for analysis:

$$\sum_{\langle i,j \rangle} \epsilon_i \epsilon_j = \epsilon M \epsilon \quad (11)$$

where M is an $N \times N$ dimensional cyclic, symmetric matrix. For our 1-D model $c_2 = c_N = 1$ and all other entries vanish

$$(M_1)_{i,j} = \delta_{i,j+1} + \delta_{i,j-1} \quad (\text{with periodic boundary conditions}) \quad (12)$$

For a 3-D lattice with n_1 sites in a row, n_2 rows in a plane, and n_3 planes, the i^{th} site is labelled by the triple (p, q, s) we assign an index to the coordinate as

$$i = p + qn_1 + sn_1n_2 \quad (13)$$

The matrix M is then given by

$$c_2 = c_{n_1+1} = c_{n_1n_2+1} = c_{N-n_1n_2+1} = c_{N-n_1+1} = c_N = 1 \quad (14)$$

and all other elements vanishing. The eigenvalues of M_d constructed in this manner (with d being the dimension of the lattice) is given by:

$$\lambda_k = 2 \sum_{i=1}^d \cos \left(\frac{2\pi n_1 n_2 \cdots n_{i-1}}{N} (k-1) \right) \quad (15)$$

where n_i is the number of lattice sites in that coordinate direction. The corresponding eigenvector is:

$$V_{ks} = N^{-1} \left[\cos \left(\frac{2\pi}{N}(k-1)(s-1) \right) + \sin \left(\frac{2\pi}{N}(k-1)(s-1) \right) \right] \quad (16)$$

We can thus make an orthogonal transformation using the eigenvalues matrix V , $\epsilon = V\mathbf{y}$

$$\epsilon M \epsilon = \sum_{k=1}^N \lambda_k y_k^2 \quad \text{and} \quad \sum \epsilon_i^2 = \sum y_i^2 \quad (17)$$

Using this orthogonal transformation, we can rewrite the ϵ integral in eq(10) as

$$\int_{\mathbb{R}^N} d^N \epsilon \exp \left\{ -s \sum \epsilon_i^2 + K \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j \right\} = \int_{\mathbb{R}^N} d^N y \exp \left\{ -\sum (s - K\lambda_i) y_i^2 \right\} \quad (18)$$

$$= \sqrt{\frac{\pi^N}{\prod_{i=1}^N (s - K\lambda_i)}} = \pi^{N/2} \exp \left\{ -\frac{1}{2} \sum \ln(s - K\lambda_i) \right\} \quad (19)$$

Note that we need $\text{Re}(s - K\lambda_i) > 0 \forall i$ so that the integral converges. Substituting this back in eq(10) and substituting $s = 2Kz$

$$Z = \frac{\Gamma(N/2)}{2(\pi)^{N/2} N^{(N-1)/2}} \pi^{N/2} 2K e^{-\frac{1}{2} N \ln 2K} \frac{1}{2\pi i} \int_{z_0-i\infty}^{z_0+i\infty} dz \left(z - \frac{1}{2} \lambda_1 \right) \exp \{ Ng(z) \} \quad (20)$$

here, $z_0 > \frac{1}{2} \lambda_{\max} = \frac{1}{2} \lambda_1$, and

$$f(z) = \lim_{N \rightarrow \infty} N^{-1} \sum_{i=2}^N \ln \left(z - \frac{1}{2} \lambda_i \right) \quad ; \quad g(z) = 2Kz = \frac{1}{2} f(z) \quad (21)$$

This integral is in the exact form used for a saddle-point approximation. Assuming z_s is a saddle-point of the integrand, the integral evaluates to

$$Z = \frac{B_N}{\sqrt{z_s - \frac{1}{2} \lambda_1} \sqrt{2\pi N \left. \frac{\partial^2 g}{\partial z^2} \right|_{z_s}}} \quad (22)$$

The saddle-point z_s is determined by $g'(z_s) = 0$ and $g''(z_s) > 0$. We have from Appendix B of Berlin and Kac[1], that

$$f_d = \frac{1}{(2\pi)^d} \int_0^{2\pi} \cdots \int_0^{2\pi} d\omega_1 \cdots d\omega_n \ln \left(z - \sum_{i=1}^d \cos \omega_i \right) \quad (23)$$

here f_d is the f function for a d dimensional lattice. Then we have

$$\frac{\partial g_d}{\partial z} = 2K - \frac{1}{2} \frac{1}{(2\pi)^d} \int_0^{2\pi} \cdots \int_0^{2\pi} d\omega_1 \cdots d\omega_n \left(z - \sum_{i=1}^d \cos \omega_i \right)^{-1} \quad (24)$$

$$\frac{\partial^2 g_d}{\partial z^2} = \frac{1}{2} \frac{1}{(2\pi)^d} \int_0^{2\pi} \cdots \int_0^{2\pi} d\omega_1 \cdots d\omega_n \left(z - \sum_{i=1}^d \cos \omega_i \right)^{-2} \quad (25)$$

$g'(z_s) = 0$ implies

$$4K = \frac{1}{(2\pi)^d} \int_0^{2\pi} \cdots \int_0^{2\pi} d\omega_1 \cdots d\omega_n \left(z_s - \sum_{i=1}^d \cos \omega_i \right)^{-1} \quad (26)$$

We get from here that z_s is real and positive. If the temperature T is very large, then K is small and hence z_s must be large. Note that the integrand has a singularity on the real axis at $z = d$ But for both the one-, and two-dimensional cases, z_s doesn't reach that singularity for any finite (non-zero) temperature. But in three dimensions we attain $z_s = 3$

for a finite temperature. This integral has been evaluated by Watson[4] to get:

$$4K_c = \frac{1}{(2\pi)^3} \int \int_0^2 \pi \int d\omega_1 d\omega_2 d\omega_3 \frac{1}{3 - \cos \omega_1 - \cos \omega_2 - \cos \omega_3} \approx 0.50546 \quad (27)$$

This means that the saddle=point exists above a critical temperature. Taking a $\ln$ of eq(22) and taking the limit $N \rightarrow \infty$ we get, for $T > T_c$

$$-\frac{\psi}{k_B T} = -\frac{1}{2} - \frac{1}{2} \ln 4K + 2Kz_s - \frac{1}{2} f_3(z_s) \quad (28)$$

We now investigate the case $T < T_c$, this requires consideration of the $z < 3$ region. Note that $f_3(z)$ has a branch cut starting at 3 and extending to $-\infty$. We look at the behaviour of $f_3(z)$ around $z = 3$. The non-analytic contribution to f_3 comes when the angles ω_i are small. We expand $\cos \omega_i = 1 - 1/2 \omega_i^2 + O(\omega_i^4)$. Therefore, the non-analytic contribution to $\partial f_3 / \partial z$ is

$$\int_{\text{small ball}} \frac{d^3 \omega}{a + \frac{1}{2} r^2} = \int_0^R \frac{r^2 dr}{a + \frac{1}{2} r^2} = a^{1/2} \cdot 2\sqrt{2} \int_0^{R/\sqrt{2a}} \frac{t^2 dt}{1 + t^2} \quad (29)$$

where $a = z - 3$ and $r^2 = \omega_1^2 + \omega_2^2 + \omega_3^2$. We see that the integral scales as $a^{1/2}$ (with a finite factor that depends on where we cut off the small-angle approximation).

$$\Rightarrow \frac{df_3}{dz} = 4K_c + C(z - 3)^{1/2} + O(z - 3) \quad (30)$$

Where the $4K_c$ term comes from the analytic part of f_3 .

$$\Rightarrow g_3(z) = g_3(3) + 2(K - K_c)(z - 3) - \frac{C}{2}(z - 3)^{3/2} + O([z - 3]^2) \quad (31)$$

For $K > K_c$ we do not get a normal saddle point. The path of steepest descent is one with a cusp at $z = 3$ as shown in fig(1), i.e, the saddle-point 'sticks' to $z = 3$. We use this information in evaluating the complex integral required for the partition function.

$$\therefore \frac{1}{2\pi i} \int_{z_0 - i\infty}^{z_0 + i\infty} dz (z - 3)^{-1/2} e^N g_3(z) \approx \frac{e^{N g_3(3)}}{N 2\pi (K - K_c)} \quad (32)$$

$$(33)$$

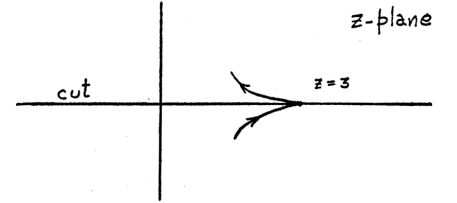


Figure 1: Contour for $T < T_c$

Hence, for $K > K_c$ ($T < T_c$)

$$-\frac{\psi}{k_B T} = -\frac{1}{2} - \frac{1}{2} \ln 4K + g_3(3) = -\frac{1}{2} - \frac{1}{2} \ln 4K + 6K - \frac{1}{2} f_3(3) \quad (34)$$

3.2 Specific Heat Capacity

Let U_n and C_n denote the internal energy per particle and specific heat per particle, respectively, for an n-dimensional simple lattice, where $n \in \{1, 2, 3\}$. We have,

$$U_n = \frac{J}{2} \frac{d}{dK} \left(\frac{\psi_n}{k_B T} \right) \text{ and } C_n = -k_B K^2 \frac{d^2}{dK^2} \left(\frac{\psi_n}{k_B T} \right) \quad (35)$$

Since, $\psi_n \equiv \psi_n(K, z_s)$

$$\Rightarrow \frac{d}{dK} \left(\frac{\psi_n}{k_B T} \right) = \frac{\partial}{\partial K} \left(\frac{\psi_n}{k_B T} \right) + \frac{dz_s}{dK} \frac{\partial}{\partial z_s} \left(\frac{\psi_n}{k_B T} \right) = \frac{\partial}{\partial K} \left(\frac{\psi_n}{k_B T} \right)$$

Where the last term in the first equality vanishes when the saddle point exists.

Specifically for the $n = 3$ case, when $T < T_c$, $z_s = 3$ and is independent of K , so the term $\frac{dz_s}{dK}$ vanishes.

Therefore,

$$\frac{d}{dK} \left(\frac{\psi_n}{k_B T} \right) = \frac{1}{2K} - 2z_s, \quad \frac{d^2}{dK^2} \left(\frac{\psi_n}{k_B T} \right) = -\frac{1}{2K^2} - 2 \frac{dz_s}{dK} \quad (36)$$

$$\Rightarrow U_n = J \left[\left(\frac{1}{4K} \right) - z_s \right] ; C_n = \frac{k}{2} \left[1 + 4K^2 \frac{dz_s}{dK} \right] \quad (37)$$

U_n is continuous since z_s is a continuous function of T . For the $n = 3$ case, the saddle-point condition $4K = \frac{df_3(z)}{dz}$, gives

$$\frac{dz_s}{dK} = 4 / \frac{d^2 f_3}{dz^2}, \quad \frac{d^2 z_s}{dK^2} = - \left(\frac{dz_s}{dK} \right)^2 \frac{d^3 f_3 / dz^3}{d^2 f_3 / dz^2}. \quad (38)$$

Near the branch point $z = 3$, where

$$\frac{d^2 f_3}{dz^2} \sim - \frac{2^{-3/2}}{\pi} (z - 3)^{-1/2}, \quad \frac{d^3 f_3}{dz^3} \sim \frac{2^{-5/2}}{\pi} (z - 3)^{-3/2}, \quad (39)$$

it follows that $\lim_{z \rightarrow 3} \frac{dz_s}{dK} = 0$ while $\lim_{z \rightarrow 3} \frac{d^2 z_s}{dK^2} = 64\pi^2$. Thus, although C_n remains continuous and approaches $\frac{1}{2}k$ as $T \rightarrow 0$, the discontinuity in $\frac{d^2 z_s}{dK^2}$ at T_c causes a break in the slope of C_3 , as shown in Fig. 2 (a detailed derivation can be found in Appendix C of [1]). This discontinuity in the slope of C_3 makes the model undergo a third order phase transition.

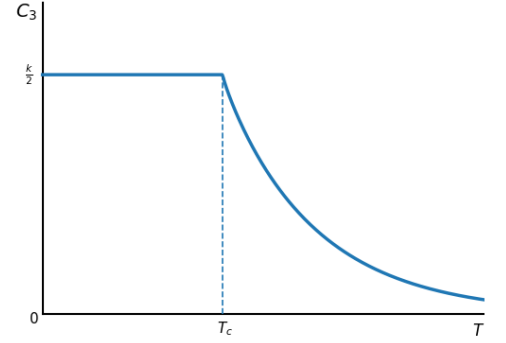


Figure 2: The specific heat per particle versus temperature for $n = 3$. There is a 3rd order phase transition at the critical point $T = T_c$.

3.3 Spontaneous Magnetization

We now proceed to derive the expression of magnetization and magnetic susceptibility from the free energy obtained in the previous sections. The average magnetic moment per particle is given by $\mu = \frac{\mu_0}{N} \sum_{j=1}^N \epsilon_j$, where μ_0 is the magnitude of magnetic moment in for each particle. Thus numerator determines the total magnetic moment of the system is a certain configuration. For net magnetization per particle of the system, unlike in Ising model, where we calculate the expectation value of the magnetic moment ($\langle \mu \rangle$) directly, we calculate the expectation value of magnitude magnetic moment. This is so because

$$\langle \mu \rangle = \langle \mu_i \rangle = N^{-\frac{1}{2}} \mu_0 \left[\frac{\frac{1}{A_N} \frac{1}{2\pi i} \int_{\alpha_0 - i\infty}^{\alpha_0 + i\infty} ds \left(e^{Ns} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} dy_1 \dots dy_N \times y_i e^{-\sum_{j=1}^N (s - K\lambda_i) y_j^2} \right)}{Z_N} \right]$$

Since the function in the inner integral is an odd function, this gives $\langle \mu \rangle = 0$. Alternatively, we can also say this results due to the symmetric nature of the system. We instead focus upon the latter.

$$\langle |\mu| \rangle = \langle |\mu_i| \rangle = N^{-\frac{1}{2}} \mu_0 \left[\frac{\frac{1}{A_N} \frac{1}{2\pi i} \int_{\alpha_0 - i\infty}^{\alpha_0 + i\infty} ds \left(e^{Ns} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} dy_1 \dots dy_N \times |y_i| e^{-\sum_{j=1}^N (s - K\lambda_i) y_j^2} \right)}{Z_N} \right]$$

(Show calculation here) When the saddle point exist for the above integration, we

$$\langle |\mu| \rangle = \frac{\mu_0}{\sqrt{N \times 2\pi \times K(z_s - \frac{1}{2}\lambda_1)}} \quad (40)$$

Note that in the above function, for the case of 1D or 2D lattices, the $z_s \rightarrow 1$ as $T \rightarrow 0$, unaffected the integral. Thus for the limit of $N \rightarrow \infty$, $\langle \mu \rangle = 0$ for both the lattice types.

4 Spherical Model in a Homogeneous Magnetic Field

The model Hamiltonian can be modified to include the effect of an external homogeneous magnetic field H as follows:

$$H = -J \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j - H \mu_0 \sum_{i=1}^N \epsilon_i = -J \sum_{\langle i,j \rangle} \epsilon_i \epsilon_j - H \mu_0 \sqrt{N} y_1 \quad (41)$$

where μ_0 is the magnetic moment associated with a single spin, and $y_1 = \frac{1}{\sqrt{N}} \sum_{i=1}^N \epsilon_i$.

In the presence of the magnetic field, the partition function becomes:

$$Z_H = \frac{1}{A_N} \frac{1}{2\pi i} \int_{-i\infty}^{i\infty} ds e^{sN} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \left(\prod_{i=1}^N dy_i \right) \exp \left(- \sum_{j=1}^N y_j^2 (s - K\lambda_j) + 2\sqrt{N} M y_1 \right) \quad (42)$$

where $M = \frac{H\mu_0}{2k_B T}$.

The y_1 integral can be evaluated by completing the square in the exponent, yielding a factor of $\exp\left(\frac{NM^2}{s-K\lambda_1}\right)$ in the integrand. The remaining integrals can be evaluated as before, as the only change is in the prefactor of s in the exponent. Following similar strategy, one can find the free energy per spin in the thermodynamic limit as:

$$\psi_H = \frac{k_B T}{2} \left(1 + \ln(4K) + 4Kz_s - f_n(z_s) + \frac{M^2}{K(z_s - \lambda_1/2)} \right) \quad (43)$$

where $z_s = z_s(K, H)$ is the solution of the saddle point equation:

$$4K = \left(\frac{df_n}{dz} \right)_{z=z_s} - \frac{M^2}{K(z_s - \lambda_1/2)^2} \quad (44)$$

For a non-zero magnetic field ($M \neq 0$), the above equation always has a saddle point when $\frac{1}{2}\lambda_1 = n$, implying there is no transition to a spontaneously magnetized state in the 3 dimensional lattice.

Using the normal saddle point, one can evaluate the mean magnetic moment per particle and magnetic susceptibility per particle as follows:

$$\langle \mu \rangle = \frac{\mu_0}{\sqrt{N}} \langle y_1 \rangle = \frac{\mu_0 M}{2K(z_s - n)} \quad ; \quad \chi = \frac{\langle \mu \rangle}{H} = \frac{\mu_0^2 H}{2J(z_s - n)} \quad (45)$$

For $n = 3$, let us define $\alpha = 6J/\mu_0^2$ and $h(T) = T_c z_s/3T$, this puts eq(45) into the form $\chi = \frac{T_c}{\alpha(h(T)T - T_c)}$, which is similar to the form of magnetic susceptibility given by the Curie-Weiss theory of ferromagnetism.

5 Spherical Model in the Literature

Ever since the original work by Berlin and Kac on the Spherical Model of a ferromagnet [1], numerous works have appeared in the literature focusing on the spherical model. Firstly, there are works that try to simplify the approach taken in the original paper by making use of constrained optimization and Lagrange multipliers such as [5] and the classic book by R. J. Baxter [6]. Henkel and Hoeger introduce the transfer-matrix approach to look at such models [7]. Some exploratory works include that of Mazo, who studied the spherical model applied to an antiferromagnetic interaction [8]. Cassie and Fabian extended the lattice spherical model to infinite graphs [9]. Works involving rigorous mathematical analysis and mathematical physics have also emerged very recently, such as [10], and more particularly the work by Lukkarinen [11] shows that the Berlin–Kac spherical model, when the particle density exceeds a critical value, the excess mass concentrates into a small number of Fourier modes (a multi-state condensation), while the remaining modes form a Gaussian “normal” background, which is a classical analogue of Bose–Einstein condensation. Lastly, some of the current research directions include investigating quantum spherical models [12] in the context of quantum information and studying the entanglement-gap/spectra of many-body quantum systems [13, 14]. Below, we very briefly introduce how we can move towards the quantum formalism [15].

Start with the classical formalism,

$$\sum_{i=1}^N S_i^2 = Nm, \quad H_{cl}[S] = -\frac{1}{2} \sum_{i,j} J_{ij} S_i S_j - \sum_i h_i S_i. \quad (46)$$

Quantize by promoting $S_i \mapsto \hat{S}_i$ and introducing conjugate momenta $\hat{\Pi}_i$ with

$$[\hat{S}_i, \hat{\Pi}_j] = i\hbar \delta_{ij}, \quad (47)$$

and enforce the spherical constraint in the quantum theory (for example in expectation value) as $\langle \sum_i \hat{S}_i^2 \rangle = Nm$, implemented with a Lagrange multiplier λ . A compact form of the quantum spherical Hamiltonian is therefore

$$\hat{H} = \frac{g}{2} \sum_i \hat{\Pi}_i^2 - \frac{1}{2} \sum_{i,j} J_{ij} \hat{S}_i \hat{S}_j - \sum_i h_i \hat{S}_i + \lambda \left(\sum_i \hat{S}_i^2 - Nm \right) \quad (48)$$

which is convenient to analyse after introducing bosonic (oscillator) operators

$$\hat{S}_i = \sqrt{\frac{\hbar}{2}} (\hat{a}_i + \hat{a}_i^\dagger), \quad \hat{\Pi}_i = \frac{1}{i} \sqrt{\frac{1}{2\hbar}} (\hat{a}_i - \hat{a}_i^\dagger), \quad [\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}, \quad (49)$$

so that $\hat{H}$ becomes quadratic in $\hat{a}, \hat{a}^\dagger$ and can be diagonalized in the eigenbasis of J_{ij} ; the multiplier λ is fixed by the saddle-point/constraint condition $\langle \sum_i \hat{S}_i^2 \rangle_T = Nm$. The classical limit is recovered by

$$\hbar \rightarrow 0 \quad \text{or} \quad g \rightarrow 0, \quad (50)$$

under which quantum kinetic fluctuations vanish and $\hat{H}$ reduces to the classical Hamiltonian $H_{\text{cl}}[S]$ together with the classical constraint $\sum_i S_i^2 = Nm$.

6 Conclusion

In summary, the spherical model of the ferromagnet is a useful framework for the examination of phase transitions and critical phenomena in statistical mechanics. Relaxing the Ising model's discrete constraints on spin and admitting continuous values of spin makes the spherical model simplify the complicated problem of ferromagnetic behavior and permit exact solutions in any dimension. From our calculations, we obtained important quantities like free energy, specific heat capacity, and spontaneous magnetization, which pointed towards the critical behavior of the system in the vicinity of the phase transition.

The addition of an external homogeneous magnetic field added further complexities, which were resolved using saddle-point approximations and perturbative methods. These additions provide insight into the evolution of magnetization and susceptibility under applied field, making the model relevant to experimental findings of real ferromagnetic systems.

In addition, the spherical model's significance in history and its persistence in contemporary physics, especially in quantum systems and research on Bose-Einstein condensation, testify to its strength and resilience. Looking into the future, the application of the spherical model to quantum systems holds out promise for future insight into quantum phase transitions and the structure of entanglement for many-body systems. The effort included here not only enhances our appreciation of classical critical phenomena but also opens the door to further investigation of quantum spherical models, bridging the divide between classical statistical mechanics and quantum mechanics.

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